Manas Ganti

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OBJECTIVE

ML Engineer with 2+ years of production experience across computer vision, edge inference, and deep learning spanning industrial AI and medical imaging. Current Computer Engineering student at Virginia Tech seeking Summer 2026 ML internship.

TECHNICAL SKILLS

Frameworks: Transformers, PyTorch, TensorFlow, Keras, OpenCV, Hugging Face, LangChain, LangGraph, CrewAI, Streamlit, ONNX, TensorRT, Nibabel, Pandas, NumPy, PEFT, FAISS, Sentence Transformers, Weights & Biases, Scikit-learn, Seaborn

Specializations: Computer Vision, NLP, Machine Learning, Deep Learning, Data Science, Neural Networks, Reinforcement Learning, Generative AI (LLMs, Agentic AI), Fine Tuning, Prompt Engineering, RAG, Model Quantization, Edge AI, Transfer Learning, Object Detection, Image Segmentation

Programming: Python, C++, SQL, R, Bash/Shell, Java, MATLAB, CUDA, JAX

Tools & Cloud: AWS, Google Cloud, Databricks, Docker, Git/GitHub, Snowflake, SQL Workbench, Kubernetes, MongoDB, MLflow

WORK EXPERIENCE

Assert AI — May 2024 – Mar 2025

Associate AI Engineer: Python, NumPy, PyTorch, DeepStream, Computer Vision

- Implemented real-time computer vision pipelines using NVIDIA **DeepStream** on Jetson Nano and Orin AGX devices, optimizing latency, GPU utilization, and throughput for industrial safety monitoring across large-scale multi-camera environments.
- Built, fine-tuned and deployed **YOLOv8** pose estimation models for hazardous zone detection, reducing false alerts to below 3% while maintaining consistent real-time inference performance under constrained memory and compute limitations on edge devices.
- Engineered a production-quality **Automatic Number Plate Recognition** (ANPR) system integrating **object detection**, **OCR**, route tracking, and post-processing logic to enable large-scale compliance and logistics monitoring across distributed industrial facilities.
- Architected and scaled a robust face detection and classification pipeline using **TensorFlow** and **transfer learning techniques**, achieving 85% accuracy on live CCTV streams under varying lighting conditions, motion blur, and partial occlusion scenarios.
- Integrated PPE compliance monitoring using custom-trained **YOLOv8** models, achieving 98% detection rates and less than 2% false positives across multi-camera real-time deployments in safety-critical industrial environments under varying lighting conditions.
- Built a centralized monitoring and analytics dashboard using **Python**, **JavaScript**, and **Seaborn** to visualize violations, track model performance metrics, and enable data-driven operational decision-making across multiple facilities using predictive analytics.

Inmed Prognostics — Oct 2022 – May 2024

AI DL Engineer: Python, NumPy, Pandas, Seaborn, OpenCV, Neural Networks, Nibabel, AntsPy, SimpleITK

- Developed, trained and scaled **UNet** architectures with **ResNet-34** backbones for brain lobe **segmentation**, achieving 95% accuracy through optimized preprocessing workflows, augmentation strategies, and structured cross-validation experiments.
- Engineered a patch-based hemorrhage detection pipeline for Brain CT scans using **ResNet-34**, improving detection sensitivity across diverse and heterogeneous data distributions while reducing inference latency to support efficient, clinical diagnostic workflows.
- Built scalable preprocessing pipelines including **Z-score normalization**, 2D slice extraction, **intensity standardization**, and augmentation techniques to improve **model generalization** and reproducibility across heterogeneous and noisy patient datasets.
- Collaborated with frontend and backend engineering teams to integrate trained deep learning and machine learning models into production-grade medical imaging systems with scalable RESTful APIs, modular architecture, and deployment-ready infrastructure.

EDUCATION

- **Master of Science in Computer Engineering**, Virginia Tech — May 2025-Dec 2027
  Relevant Coursework: Advanced Machine Learning, Computer Vision, Human Robot Interaction, Reinforcement Learning, Computer Architecture, Applications of Machine Learning
- **Bachelor of Engineering in Electronics and Communications,** MIT WPU — Jun 2018 - May 2022
  Relevant Coursework: Fuzzy Logic, Data Structures and Algorithms, Machine Learning, Digital Electronics, Signals and Systems, Digital Signal Processing, Linear Integrated Circuits

PROJECTS

Real Time Soccer Player Tracking and Analysis system using OpenCV and Pytorch

Tech stack: Python, PyTorch, OpenCV, Roboflow

- Integrated a real-time multi-object tracking pipeline using **OpenCV** and **PyTorch** to detect, track player-level motion metrics.
- Built a custom team-classification module using **SigLIP** embeddings with **UMAP** and **KNN** clustering for robust player grouping.
- Designed a preprocessing pipeline for transfer learning on YOLOv8n model, achieving 88% detection accuracy and optimized inference performance for frame-by-frame processing under real-time constraints and calculating player's current and average speeds.

Detecting artefacts in Brain MRI scans using 3D Classification networks

Tech stack: Python, PyTorch, NumPy, Weights & Biases, Nibabel, Keras, Tensorflow

- Implemented a custom 3D classifier network to differentiate between regular brain MRI scans and artefact affected MRI scans.
- Trained on a custom dataset using PyTorch; augmented datasets, tuned optimizer, learning rate and scheduler to achieve an **85% accuracy** on random sampled brain MRI scans.
- Analyzed model with *real time loss* function curves and visualized model performance using **Matplotlib**

Medical Report Generation using Fine Tuned LLMs on MIMIC-CXR Dataset

Tech stack: Python, PyTorch, LLaMA-3, Hugging Face, PEFT/QLoRA, FAISS, LangChain, Weights & Biases, MIMIC-CXR

- Fine-tuned LLaMA-3 8B with QLoRA/PEFT on the MIMIC-CXR radiology dataset to generate structured diagnostic summaries from Brain CT/MRI metadata and segmentation model outputs; reduced GPU memory requirements by 4x vs. full fine-tuning using 4-bit quantization.
- Designed a RAG pipeline with FAISS vector store and Sentence Transformers for context-grounded generation, grounding outputs in patient history and reducing hallucinated findings vs. non-RAG baseline.
- Achieved ROUGE-L of 0.43 and BERT Score F1 of 0.87 on held-out radiology reports logged all training runs and loss curves using Weights and Biases.